European Journal of Cardio-Thoracic Surgery 2025, 67(1), ezae414
https://doi.org/10.1093/ejcts/ezae414 Advance Access publication 20 November 2024

ORIGINAL ARTICLE

CONVENTIONAL VALVE OPERATIONS

Cite this article as: Sabik III JF, Rao V, Dagenais F, Moront MG, Reardon MJ, Patel HJ *et al*. Seven-year outcomes after surgical aortic valve replacement with a stented bovine pericardial bioprosthesis in over 1100 patients: a prospective multicentre analysis. Eur J Cardiothorac Surg 2025; doi:10.1093/ejcts/ezae414.

Seven-year outcomes after surgical aortic valve replacement with a stented bovine pericardial bioprosthesis in over 1100 patients: a prospective multicentre analysis

Joseph F. Sabik III[a,*], Vivek Rao [b], Francois Dagenais[c], Michael G. Moront[d], Michael J. Reardon [e], Himanshu J. Patel [f], Jae K. Oh[g], Shinichi Fukuhara[f], Louis Labrousse [h], Ralf Günzinger[i], Kamran Baig[j], Saki Ito [g], Tianhua Wu[k] and Robert J. M. Klautz[l], on behalf of the PERIGON Investigators

[a]Department of Surgery, University Hospitals Cleveland Medical Center and Case Western Reserve University School of Medicine, Cleveland, OH, USA
[b]Peter Munk Cardiac Centre, Toronto General Hospital, Toronto, ON, Canada
[c]Cardiac Surgery Service, Quebec Heart and Lung Institute, Quebec, QC, Canada
[d]Department of Cardiothoracic Surgery, ProMedica Toledo Hospital, Toledo, OH, USA
[e]Department of Cardiovascular Surgery and Cardiology, Houston Methodist DeBakey Heart and Vascular Center, Houston, TX, USA
[f]Department of Cardiac Surgery, University of Michigan, Ann Arbor, MI, USA
[g]Department of Cardiovascular Medicine, Mayo Clinic, Rochester, MN, USA
[h]Medico-Surgical Department of Valvulopathies, Bordeaux Heart University Hospital, Bordeaux-Pessac, France
[i]Department of Cardiovascular Surgery, Institute Insure, German Heart Center Munich, School of Medicine & Health, Technical University of Munich, Munich, Germany
[j]Cardiovascular Department, Guys & St Thomas NHS Foundation Trust–St Thomas Hospital, London, UK
[k]Clinical Research and Medical Science, Cardiac Surgery, Medtronic, Mounds View, MN, USA
[l]Department of Cardiothoracic Surgery, Leiden University Medical Center, Leiden, The Netherlands

* Corresponding author. Department of Surgery, University Hospitals Cleveland Medical Center, Case Western Reserve University, 11100 Euclid Avenue, Lakeside 7, Cleveland, OH 44106-7060, USA. Tel: +1-216-844-3211; e-mail: Joseph.Sabik@Uhhospitals.org (J.F. Sabik III).

Received 31 July 2024; received in revised form 25 October 2024; accepted 18 November 2024

7-Year outcomes after SAVR with a stented bovine pericardial bioprosthesis in >1100 patients: a prospective multicenter analysis

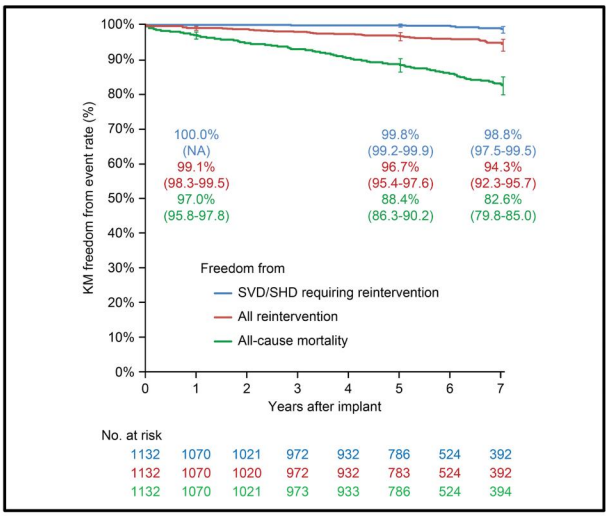
Summary

Population: Patients with moderate to severe AS or severe AR.

Intervention: SAVR with a stented bovine pericardial bioprosthesis.

Comparison: Clinical outcomes, durability, hemodynamic performance, and functional status 7 years after AVR.

Outcomes: 83% survival, 94% freedom from all reintervention, and 99% freedom from SVD/SHD requiring reintervention at 7 years of follow-up.



Legend: AR, aortic regurgitation; AS, aortic stenosis; SAVR, surgical aortic valve replacement; SHD, severe hemodynamic dysfunction; SVD, structural valve deterioration.

Presented at the 38th EACTS Annual Meeting, Lisbon, Portugal, 9–12 October 2024.

© The Author(s) 2024. Published by Oxford University Press on behalf of the European Association for Cardio-Thoracic Surgery.
This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial License (https://creativecommons.org/licenses/by-nc/4.0/), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited. For commercial re-use, please contact journals.permissions@oup.com

Abstract

OBJECTIVES: Safety, efficacy and durability are important considerations when selecting a bioprosthesis for aortic valve replacement (AVR). This study assessed 7-year clinical outcomes and haemodynamic performance of the Avalus bioprosthesis.

METHODS: Patients indicated for surgical AVR were enrolled in this prospective, nonrandomized trial, conducted across 39 sites globally. The primary end-point of this analysis was freedom from surgical explant or percutaneous valve-in-valve reintervention due to structural valve deterioration (SVD) at 7 years of follow-up, determined using Kaplan–Meier (KM) analysis. We also evaluated a composite end-point of SVD and/or severe haemodynamic dysfunction requiring reintervention. Survival, valve-related safety events and haemodynamic performance were assessed. Deaths and safety events were adjudicated by an independent clinical events committee.

RESULTS: A total of 1132 patients underwent surgical AVR. Mean age was 70 years; 854 patients (75%) were men. The mean STS risk of mortality was 2.0 ± 1.4%, and 659 patients (58%) had a New York Heart Association classification of I/II. One or more concomitant procedures were performed in 577 patients (51%). At 7 years, the Kaplan–Meier rate of freedom from SVD/severe haemodynamic dysfunction requiring reintervention was 1.2% (0.5–2.5%) with no cases adjudicated as SVD. The survival rate was 82.6% (79.5–85.0%). The KM event rate was 5.7% (4.3–7.7%) for reintervention, 6.3% (4.9–8.3%) for endocarditis and 0.4% (0.1–1.1%) for valve thrombosis. Mean aortic gradient, dimensionless velocity index and effective orifice area were 13.8 ± 5.9 mmHg, 0.42 ± 0.09 and 1.99 ± 0.53 cm^2, respectively.

CONCLUSIONS: This analysis demonstrated excellent durability of the Avalus valve with good clinical outcomes and stable haemodynamic performance through 7 years of follow-up.

Keywords: Surgical aortic valve replacement • Durability • Hemodynamic performance

ABBREVIATIONS

AS	Aortic stenosis
DVI	Dimensionless velocity index
EOA	Effective orifice area
LTFU	Long-term follow-up
NSVD	Non-structural valve dysfunction
NYHA	New York Heart Association
PERIGON	PERIcardial SurGical AOrtic Valve ReplacemeNt clinical trial of the Avalus valve
PVL	Paravalvular leak
SAVR	Surgical aortic valve replacement
SHD	Severe hemodynamic dysfunction
SVD	Structural valve deterioration
TVL	Transvalvular leak

INTRODUCTION

Bioprosthetic valves have been widely used for surgical aortic valve replacement (SAVR) because lifelong anticoagulation is not required, and they pose a much lower risk of bleeding and thrombotic complications than mechanical valves. However, durability is a consideration because the tissue used in bioprosthetic valves is prone to deterioration over time.

The intermediate-term follow-up period, defined as ∼5–7 years, is a critical period for evaluating the durability and haemodynamic stability of newly introduced bioprosthetic valves. Recently, the Trifecta valve was removed from commercial distribution due to concerns about durability, which became apparent predominantly between 5 and 8 years of follow-up [1]. Similarly, the Toronto SPV valve was also removed from commercial distribution because of a higher-than-expected rate of failure due to central regurgitation, a trend that became apparent at ∼7 years [2]. The PERIcardial SurGical AOrtic Valve ReplacemeNt (PERIGON) Pivotal Trial (Medtronic, Minneapolis, MN, USA) evaluated clinical outcomes and haemodynamic performance of the Avalus valve, a stented pericardial valve with an amino-oleic-acid anticalcification treatment. This analysis evaluated outcomes through 7 years.

PATIENTS AND METHODS

Ethical statement

The institutional review board or ethics committee (IRB/EC) at all sites approved the protocol (www.clinicaltrials.gov, NCT02088554; Supplementary Material, Table S1). All patients provided written informed consent. The pivotal trial had a planned duration of 5 years, which was extended to 12 years to continue data collection. Sites with 10 or more active patients at the time of evaluation were invited to continue in long-term follow-up (LTFU); 19 sites agreed to participate. An addendum to the original protocol for the pivotal study was sent to the IRB/EC at each participating site; additional approval numbers were issued if required (Supplementary Material, Table S1). All active patients at the LTFU sites were invited to continue in follow-up. Patients who accepted provided new written informed consent, and 576 patients were re-consented. At the time of analysis, 441 of 458 patients (96%) with an expected 7-year visit had completed the visit.

Study design

The PERIGON Pivotal Trial is a prospective, multicentre trial conducted at 39 sites in Europe, Canada and the United States. The trial, previously described in detail [3, 4], enrolled adults with moderate or severe aortic stenosis (AS) and/or chronic severe aortic regurgitation and an indication for SAVR. In the pivotal trial, in-person follow-up was performed at 3–6 months, 1 year and then annually through 5 years. For the LTFU study, follow-up will be conducted in person at years 7, 10 and 12 and by telephone at years 6, 8, 9 and 11. The following evaluations will be conducted annually: New York Heart Association (NYHA) classification, medication use (i.e. aspirin, other antiplatelet, anticoagulant and indication), vital status, and safety end-points as

defined in the primary protocol. Transthoracic echocardiography and 12-lead electrocardiography will be performed at the 7-, 10- and 12-year visits.

End-points

The primary end-point of this analysis was freedom from surgical explant and/or percutaneous valve-in-valve reintervention due to structural valve deterioration (SVD) through 7 years of follow-up. Originally, in the PERIGON Pivotal Trial, SVD was defined as 'Change to the function of a heart valve substitute resulting from an intrinsic abnormality that causes stenosis or regurgitation. The diagnosis should be confirmed by examination of the explanted or damaged valve. It includes intrinsic changes such as wear, fatigue failure, stress fracture, calcification, leaflet tear and stent creep. This definition excludes paravalvular leak (PVL), infection, pannus overgrowth or thrombosis of the heart valve substitute as determined by reoperation, autopsy, or *in vivo* investigation' [5]. We also analyzed a composite end-point of SVD requiring reintervention and/or severe haemodynamic dysfunction (SHD) requiring reintervention to address the limitations of the protocol definition of SVD. The introduction of TAV-in-SAV as a reintervention method for valve failure has made it difficult to definitively determine the failure mode in some patients with significant haemodynamic dysfunction. The SHD end-point was used to categorize potential safety events that were characterized by severe AS, severe transvalvular regurgitation or progressive severe dysfunction that did not meet the protocol definitions of SVD or non-structural valve dysfunction (NSVD) [5]. We also report the Kaplan–Meier (KM) event rates for the individual components of the composite end-point.

Secondary end-points were the KM rates of survival (freedom from all-cause mortality), freedom from valve-related reintervention, freedom from valve-related death, NYHA functional classification status, and haemodynamic performance, including mean aortic gradient, dimensionless velocity index (DVI), effective orifice area (EOA), PVL, transvalvular leak (TVL) and prosthesis-patient mismatch (PPM). DVI is the ratio between the velocity time integral (VTI) of the left ventricular outflow track (LVOT) and of the aortic prosthesis (AV), i.e. LVOT VTI/AV VTI.

Valve-related safety end-points included valve-related thromboembolism, valve thrombosis, haemorrhage, PVL, endocarditis, hemolysis, NSVD, SVD, reintervention and explant [6, 7]. The safety end-points were also stratified by surgical procedure, i.e. isolated SAVR, SAVR + coronary artery bypass graft and SAVR + other concomitant procedures.

Deaths and valve-related safety events were adjudicated by an independent clinical events committee (Baim Clinical Research Institute, Boston, MA, USA), and explanted valves were evaluated by CV Path Institute (Gaithersburg, MD, USA). Echocardiograms obtained at the 3- to 6-month, 1-year, 5-year and 7-year visits were assessed by a core laboratory (Mayo Clinic, Rochester, MN, USA).

Statistical analysis

Categorical data are reported as frequencies and percentages, and continuous data are reported as means and standard deviations (SDs). KM analysis was performed to determine freedom from surgical explant and/or percutaneous valve-in-valve reintervention due to SVD/SHD requiring reintervention, freedom from valve-related reintervention, freedom from valve-related death and survival at 7 years. KM analysis was also used to calculate the event rates of valve-related safety end-points and the composite end-point. Haemodynamic performance was evaluated by analyzing the means and SDs of aortic gradient, DVI and EOA, as well as the severity of PVL and TVL.

Analyses were performed with SAS version 9.4 (SAS Institute, Cary, NC, USA) and R version 4.3.2 (The R Project for Statistical Computing, www.r-project.org).

RESULTS

Baseline and procedural characteristics

A total of 1312 patients were enrolled, and 1281 underwent baseline evaluation. In total, 1132 underwent successful implantation and are included in this analysis. Total follow-up for the full cohort was 6412 patient-years, and late follow-up (>30 days) was 6319 patient-years. Supplementary Material, Fig. S1 provides more information about patient follow-up and exits from the study.

The results reported in this manuscript are for the full cohort. Mean age at baseline was 70 years with 76% of patients >65 years of age (Supplementary Material, Fig. S2). Seventy-five percentage of patients were male. Fifty-eight % had a NYHA functional status of class I or II, and the mean Society of Thoracic Surgeons (STS) predicted risk of mortality was 2.0%. Hypertension, dyslipidaemia, and coronary artery disease were the most frequent comorbidities. Table 1 provides additional patient characteristics.

The primary indication for SAVR was AS in 84% of patients, aortic regurgitation in 6% of patients, mixed AS/aortic regurgitation in 10% of patients and a failed prosthesis in <1% of patients. Thirty % of patients had a congenital bicuspid valve. A median sternotomy was the most frequent surgical approach (80%), followed by hemisternotomy (13%), right thoracotomy (6%) and 'other' approach (1%). Coronary artery bypass graft was the most common concomitant procedure, performed in 32% of patients. The most common valve size was 23 mm, implanted in 35% of patients (Supplementary Material, Fig. S3). Table 2 provides additional procedural details.

End-points

Freedom from surgical explant and/or percutaneous valve-in-valve reintervention due to SVD/SHD at 7 years was 98.8% (97.5–99.5%) ($n = 7$). All 7 cases were adjudicated as SHD requiring reintervention, and none were adjudicated as SVD. Because of the limitations of the protocol definition, the criteria for SVD were not met for these 7 cases, and it was not possible to determine the precise mode of failure. Thus, these cases were adjudicated as SHD. Supplementary Material, Table S2 provides additional details about these 7 patients. Freedom from valve-related reintervention was 94.3% (92.3–95.7%) at 7 years. Overall survival (freedom from all-cause mortality) was 82.6% (79.8–85.0%), and freedom from valve-related death was 97.4% (95.9–98.3%) (Fig. 1).

Table 3 shows the KM rates of valve-related safety events. At 7 years, the KM event rate was 6.3% (4.9–8.1%) for thromboembolism, 0.4% (0.1–1.1%) for valve thrombosis, 9.7% (8.1–11.6%)

Table 1: Baseline characteristics

Characteristic	Patients $N = 1132$
Age (years)	70.1 ± 8.9
Male sex	854 (75.4%)
Body surface area (m^2)	2.0 ± 0.2
NYHA class	
I	126 (11.1%)
II	533 (47.1%)
III	451 (39.8%)
IV	22 (1.9%)
STS risk of mortality (%)	2.0 ± 1.4
History of atrial fibrillation	119/1125 (10.6%)
Paroxysmal	72/1125 (6.4%)
Persistent	32/1125 (2.8%)
Long-standing persistent	15/1125 (1.3%)
Unknown	7
Congestive heart failure	224 (19.8%)
Coronary artery disease	496 (43.8%)
Diabetes	303 (26.8%)
Dyslipidaemia	698 (61.7%)
Prior endocarditis	4 (0.4%)
Hypertension	861 (76.1%)
Left ventricular hypertrophy	459 (40.5%)
Myocardial infarction	100 (8.8%)
Renal dysfunction/insufficiency	120 (10.6%)
Stroke/cerebrovascular accident	46 (4.1%)
Transient ischaemic attack	61 (5.4%)
Coronary artery bypass	25 (2.2%)
Percutaneous coronary intervention	161 (14.2%)
Stent implanted	153 (13.5%)
Implanted cardiac device (i.e. pacemaker or defibrillator)	39 (3.4%)
Previous open-heart surgery[a]	40 (3.5%)

Data are mean ± SD or *n* (%).
[a]No patient had >1 prior open-heart surgery.
NYHA: New York Heart Association; SD: standard deviation; STS: Society of Thoracic Surgeons.

Table 2: Procedural characteristics

Characteristic	Patients $N = 1132$
Primary indication for valve replacement	
Aortic stenosis	954 (84.3%)
Aortic regurgitation	65 (5.7%)
Mixed	107 (9.5%)
Failed prosthesis	6 (0.5%)
Surgical approach	
Median sternotomy	903 (79.8%)
Hemisternotomy	146 (12.9%)
Right thoracotomy	69 (6.1%)
Other	14 (1.2%)
Concomitant procedures[a]	
None	555 (49.0%)
Coronary artery bypass graft	365 (32.2%)
Implantable cardiac device[b]	1 (0.1%)
Left atrial appendage closure	91 (8.0%)
Patent foramen ovale closure	13 (1.1%)
Resection of subaortic membrane not requiring myectomy	21 (1.9%)
Ascending aortic aneurysm repair not requiring circulatory arrest	92 (8.1%)
Dissection repair not requiring circulatory arrest	1 (0.1%)
Other	161 (14.2%)
Annular enlargement[c]	33/614 (5.4%)
Aortic root/STJ enlargement[c]	90/617 (14.6)
Total bypass time, min	105.6 ± 41.4
Total aortic cross-clamp time (min)	79.4 ± 31.3
Congenital bicuspid native aortic valve	338 (29.9%)

[a]Patients may have had >1 procedure.
[b]Pacemaker, implantable cardiac defibrillator, coronary revascularization therapy, etc.
[c]This procedure was not collected on the initial case report form but was added later.

for all haemorrhage, 1.0% (0.5–1.8%) for PVL, 6.3% (4.9–8.3%) for endocarditis, 5.7% (4.3–7.7%) for reintervention and 4.4% (3.1–6.0%) for explant. Supplementary Material, Fig. S4 shows the rates of thromboembolism, endocarditis, haemorrhage, and reintervention when analyzed with death as a competing outcome. The KM event rate at 7 years was 2.8% (1.7–4.6%) for all SHD (requiring/not requiring reintervention) and 1.2% (0.5–2.5%) for SVD/SHD requiring reintervention. The rates for the subcategories were, respectively, 0% (0–0%) and 1.2 (0.5–2.5%). There were no adjudicated cases of SVD not requiring reintervention. At 7 years, the KM rates of valve-related safety events were similar among all three surgical procedure groups except for all haemorrhage, which was higher in the SAVR + 'other' concomitant procedures group (14.0% vs 7.9% vs 8.2%, $P = 0.019$; Supplementary Material, Table S3).

Forty-seven patients underwent a reintervention by the time of analysis. Of the 38 patients who had a surgical reintervention (explant), 33 had endocarditis, 1 had valve thrombosis, 1 had major PVL and 2 had NSVD (1 major PVL, 1 entrapment by pannus, tissue, or suture). Among the 9 patients who had a percutaneous reintervention, 7 had SHD, 1 had valve thrombosis, 1 had major PVL and 1 had NSVD (major PVL). Figure 2 shows the KM rates of reintervention stratified by cause. There were no statistical differences in the rates of valve-related events that led to a reintervention in patients ≤65 and >65 years of age (Supplementary Material, Table S4).

At 7 years, the mean aortic gradient was 13.8 ± 5.9 mmHg ($n = 409$), DVI was 0.42 ± 0.09 ($n = 388$), and EOA was 1.99 ± 0.53 ($n = 367$) (Supplementary Material, Table S5). Supplementary Material, Fig. S5 illustrates 99.0% and 98.3% of patients had none or trace PVL and TVL at 7 years, respectively. At 7 years, 76.8% of patients had no PPM, 19.6% had moderate PPM, and 3.5% had severe PPM (Supplementary Material, Fig. S6). The proportion of patients with an NYHA functional classification of I/II was ≥94% after valve implantation through 7 years of follow-up (Supplementary Material, Fig. S7).

DISCUSSION

This analysis of the PERIGON trial demonstrates that the Avalus valve delivers excellent clinical outcomes. Over 7 years of follow-up, echocardiographic and clinical data reveal consistently stable haemodynamic performance and functional status. These findings underscore the valve's reliability and efficacy. Freedom from SVD/SHD requiring reintervention at 7 years was 98.8% (97.5–99.5%), indicating excellent intermediate durability of the study valve. Survival from all causes was 82.6% (79.8–85.0%). The mean aortic gradient was 13.8 ± 5.9 mmHg, EOA was 1.99 ± 0.53 cm^2 and DVI was 0.42 ± 0.09 at 7 years of follow-up. Rates of none/trace PVL and TVL were 99.0% and 98.3%, respectively. Functional status has been stable with ≥94% of patients in NYHA class I or II throughout follow-up.

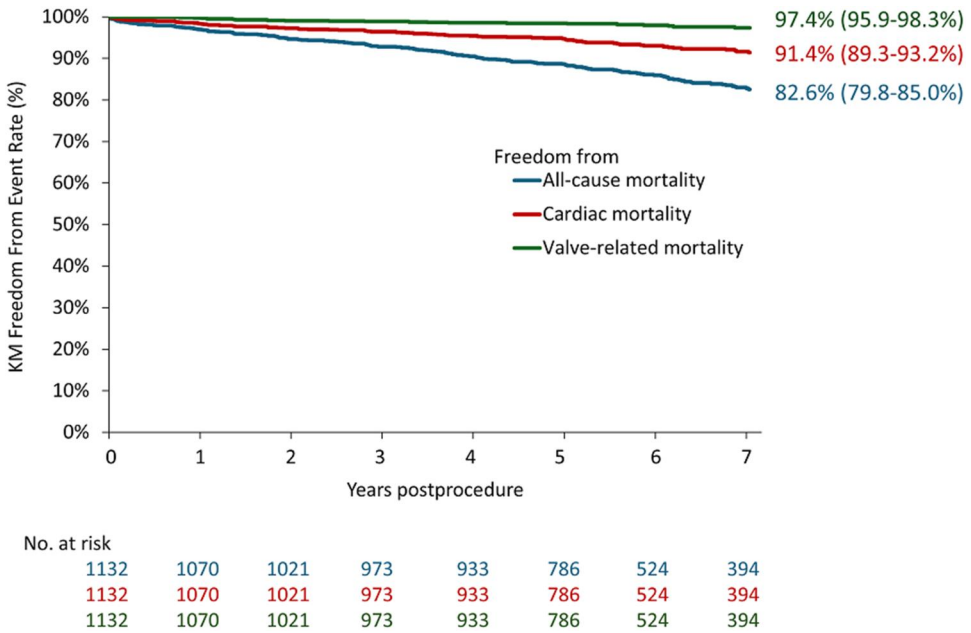


Figure 1: Survival (KM event rates of freedom from all-cause, cardiac and valve-related mortality) through 7 years of follow-up. KM: Kaplan–Meier.

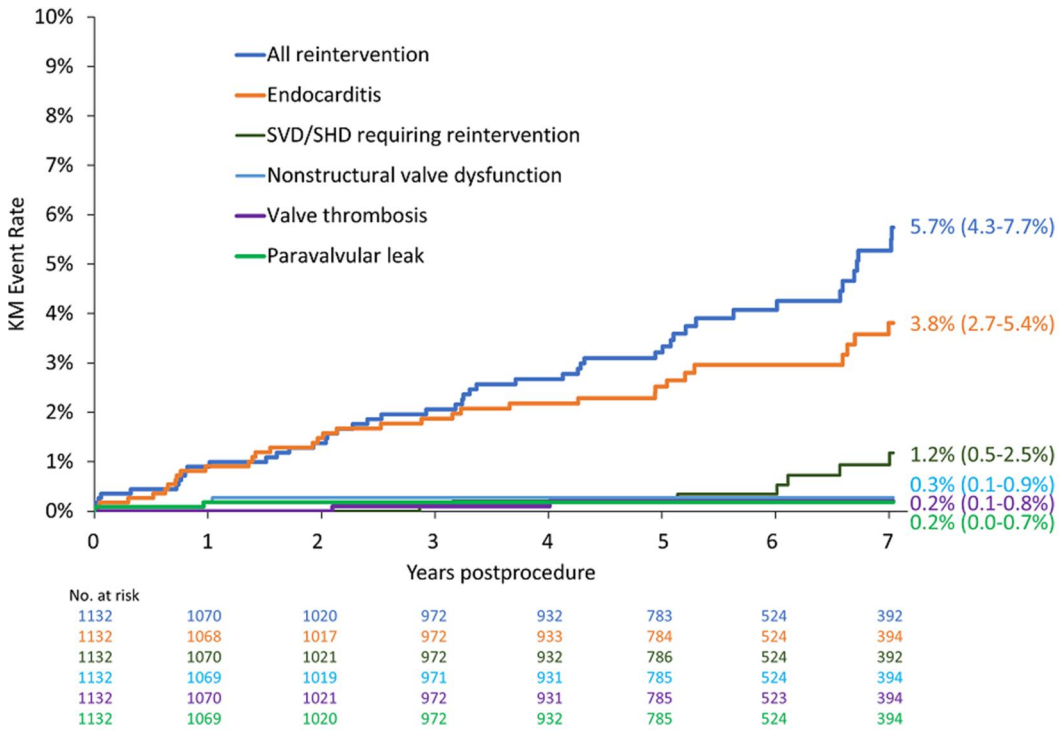


Figure 2: KM rate of reintervention (surgical or percutaneous) stratified by cause through 7 years of follow-up. KM: Kaplan–Meier.

The primary end-point for this analysis was defined as freedom from surgical explant and/or percutaneous valve-in-valve reintervention due to SVD. We also evaluated a composite end-point of SVD/SHD requiring reintervention to address the limitations of the protocol definition of SVD. While all 7 cases of this composite end-point were adjudicated as SHD by the clinical events committee, this does not definitively rule out the presence of SVD in the study cohort. Traditional surgical definitions of SVD have limitations, particularly with the increased use of valve-in-valve procedures, making it challenging to distinguish between thrombus, NSVD and SVD using *in situ* imaging for diagnosing valvular dysfunction. Although these cases might be considered SVD on the basis of haemodynamic criteria, such as those recommended by the Valve Academic Research Consortium 3 [8], a recent publication described the limitations of those definitions as well [9]. Reliable and standardized

Table 3: Kaplan–Meier rates of death and valve-related safety events at 30 days and 1, 5 and 7 years of follow-up[a]

Event	Event rate, %	Kaplan–Meier event rate, % (95% CI)[b]		
	30 days	1 year	5 years	7 years
All death	0.9 $n=10$	3.0 (2.2–4.2) $n=34$	11.6 (9.8–13.7) $n=124$	17.4 (15.0–20.2) $n=160$
Cardiac death	0.5 $n=6$	1.6 (1.0–2.6) $n=18$	5.5 (4.3–7.1) $n=57$	8.6 (6.8–10.7) $n=75$
Valve-related death	0.0 $n=0$	0.3 (0.1–0.9) $n=3$	1.6 (1.0–2.5) $n=16$	2.6 (1.7–4.1) $n=22$
Thromboembolism	1.4 $n=15$	2.7 (1.9–3.8) $n=30$	5.6 (4.4–7.2) $n=59$	6.3 (4.9–8.1) $n=63$
Valve thrombosis	0.0 $n=0$	0.0 (0.0–0.0) $n=0$	0.4 (0.1–1.1) $n=4$	0.4 (0.1–1.1) $n=4$
All haemorrhage[c]	1.6 $n=17$	5.1 (4.0–6.6) $n=57$	9.7 (8.1–11.6) $n=103$	9.7 (8.1–11.6) $n=103$
Major haemorrhage[c]	1.0 $n=11$	3.4 (2.5–4.7) $n=38$	5.8 (4.5–7.4) $n=62$	5.8 (4.5–7.4) $n=62$
All paravalvular leak	0.2 $n=2$	0.5 (0.2–1.2) $n=6$	1.0 (0.5–1.8) $n=10$	1.0 (0.5–1.8) $n=10$
Major paravalvular leak	0.1 $n=1$	0.2 (0.0–0.7) $n=2$	0.2 (0.0–0.7) $n=2$	0.2 (0.0–0.7) $n=2$
Endocarditis	0.2 $n=2$	1.2 (0.7–2.0) $n=13$	4.4 (3.3–5.8) $n=45$	6.3 (4.9–8.3) $n=56$
Hemolysis	0.0 $n=0$	0.0 (0.0–0.0) $n=0$	1.1 (0.6–2.0) $n=10$	1.3 (0.7–2.3) $n=11$
Non-structural valve dysfunction	0.2 $n=2$	0.5 (0.2–1.2) $n=6$	1.3 (0.7–2.2) $n=13$	1.3 (0.7–2.2) $n=13$
SVD/SHD requiring reintervention	0.0 $n=0$	0.0 (0.0–0.0) $n=0$	0.2 (0.1–0.8) $n=2$	1.2 (0.5–2.5) $n=7$
SHD requiring reintervention	0.0 $n=0$	0.0 (0.0–0.0) $n=0$	0.2 (0.1–0.8) $n=2$	1.2 (0.5–2.5) $n=7$
SVD requiring reintervention[d]	0.0 $n=0$	0.0 (0.0–0.0) $n=0$	0.0 (0.0–0.0) $n=0$	0.0 (0.0–0.0) $n=0$
Reintervention	0.4 $n=4$	0.9 (0.5–1.7) $n=10$	3.3 (2.4–4.6) $n=34$	5.7 (4.3–7.7) $n=47$
Explant	0.4 $n=4$	0.9 (0.5–1.7) $n=10$	2.9 (2.1–4.2) $n=30$	4.4 (3.1–6.0) $n=38$

[a]Analysis cohort is all implanted patients ($N=1132$).
[b]Patients may have had >1 event.
[c]Anticoagulant-related haemorrhage only.
[d]There were no adjudicated cases of SVD requiring or not requiring reintervention.
SHD: severe hemodynamic dysfunction; SVD: structural valve deterioration.

methods to identify SVD in both surgical and transcatheter valve clinical trials seem therefore warranted. We believe a reasonable approach is to remain agnostic regarding the mechanism of valvular dysfunction when unknown, but to conservatively group such events with SVD for comparison to historical and contemporary event rates from other trials.

The clinical and haemodynamic outcomes reported here are comparable to those reported by other studies of SAVR. Beaver *et al.* [10] reported a freedom from SVD rate of 99.3% at 7 years in the COMMENCE trial, whereas Tsui *et al.* [11] reported a freedom from reintervention due to an SVD rate of 93.6% at 8 years (median follow-up = 7.0 years). In contrast, studies of the Trifecta and Mitroflow prostheses have demonstrated that the incidence of SVD begins to increase substantially after 6 years [12, 13]. In this study, freedom from valve-related reintervention at 7 years was 94.3%, whereas Beaver *et al.* [10] reported a freedom from reoperation rate of 97.2%, and Tsui *et al.* [11] reported that freedom from reintervention at 8 years was 89.8%. Overall survival at 7 years in the PERIGON trial was 82.6%, compared with 85.4% and 80.7% reported by others [10, 11]. Freedom from valve-related death was 97.4% at 7 years in PERIGON, compared with 95.8% at 8 years in the Tsui et al. study [11].

Mean gradient and EOA, in addition to PVL and TVL, were similar to what has been reported for other surgical aortic valves [10, 11]. DVI was 0.42 ± 0.09 in our study. This index is useful for evaluating valve stenosis because it is not dependent on blood flow and does not require measurement of the left ventricular outflow tract. A DVI >0.30–0.35 is considered normal for a prosthetic valve [14, 15]. However, discharge DVIs ≤0.50 and ≤0.35 have been associated with worse clinical outcomes [16, 17].

Limitations

While the study does face some limitations (such as patient attrition after the 5-year visit), with 576 of the 1132 patients reconsenting to LTFU, it remains robust. The potential for selection bias is minimal, as only sites with 10 or more active patients at the 5-year evaluation were invited to participate. Although there is no control group to compare different valves, the study is the first to provide 7-year outcomes for the Avalus valve. Although echo follow-ups are not annual, with echocardiograms taken at 3- to 6-month, 1-, 5- and 7-year visits, these intervals still provide valuable insights. Finally, while the SVD end-point is based on the

traditional surgical definition, it remains a valid comparison point alongside studies using longitudinal haemodynamic data criteria.

CONCLUSION

The 7-year data reported here demonstrate excellent durability of the Avalus valve with continued low rates of SVD/SHD requiring reintervention, mortality, and valve-related reinterventions. These results are important to guide clinical decision-making in the lifetime management of aortic valve disease.

SUPPLEMENTARY MATERIAL

Supplementary material is available at *EJCTS* online.

ACKNOWLEDGEMENTS

Julie A. Linick, of Medtronic, assisted in the drafting and revision of this manuscript under the direction of the authors. Shuzhen Li, of Medtronic, assisted with the statistical analysis.

FUNDING

Funded by Medtronic.

Conflict of interest: Joseph F. Sabik III: North American Principal Investigator of the PERIGON Pivotal Trial, Medtronic. Vivek Rao: Medtronic: Consultant, North American Surgical Advisory Board, Equity (<$25k). Abbott: Consultant. Gore: Consultant. Francois Dagenais: Honoraria from Medtronic, Abbott and Edwards for presentations and from Cook Medical for proctoring and presentation. Michael G. Moront: Trainer and consultant, Medtronic; trainer and speaker, Atricure; speaker and consultant, Haemonetics. Michael J. Reardon: Consultant to Medtronic; all payments go directly to his department. Himanshu J. Patel: Consultant and Surgical Advisory Board member for Medtronic. Jae K. Oh: Director of the Echo Core Lab for the PERIGON LTFU trial; consulting for valve diseases, Medtronic; Anumana equity for ECG AI for aortic stenosis. Shinichi Fukuhara: Consultant for Edwards Lifesciences, Medtronic, Artivion and Terumo Aortic. Tianhua Wu: Medtronic employee. Robert J.M. Klautz: European Principal Investigator of the PERIGON Pivotal Trial, research support and consultation fees from Medtronic. Louis Labrousse, Ralf Günzinger, Kamran Baig, and Saki Ito: none.

DATA AVAILABILITY

The data underlying this article are owned by the study sponsor and will not be made available to other investigators for research purposes.

Author contributions

Joseph F. Sabik III: Conceptualization; Investigation; Resources; Supervision; Visualization; Writing—original draft. **Vivek Rao:** Investigation; Resources; Writing—review & editing. **Francois Dagenais:** Investigation; Resources; Writing—review & editing. **Michael G. Moront:** Investigation; Resources; Writing—review & editing. **Michael J. Reardon:** Investigation; Resources; Writing—review & editing. **Himanshu J. Patel:** Investigation; Resources; Writing—review & editing. **Jae K. Oh:** Investigation; Writing—review & editing. **Shinichi Fukuhara:** Investigation; Resources; Writing—review & editing. **Louis Labrousse:** Investigation; Resources; Writing—review & editing. **Ralf Günzinger:** Investigation; Resources; Writing—review & editing. **Kamran Baig:** Investigation; Resources; Writing—review & editing. **Saki Ito:** Investigation; Writing—review & editing. **Tianhua Wu:** Formal analysis; Methodology; Writing—review & editing. **Robert J.M. Klautz:** Conceptualization; Investigation; Resources; Supervision; Writing—review & editing.

Reviewer information

European Journal of Cardio-Thoracic Surgery thanks Mateo Marin-Cuartas and the other anonymous reviewers for their contribution to the peer review process of this article.

REFERENCES

[1] US Food and Drug Administration. Abbott Trifecta Valves: Potential Risk of Early Structural Valve Deterioration—Letter to Health Care Providers. Silver Spring, MD, 2023. https://www.fda.gov/medical-devices/letters-health-care-providers/abbott-trifecta-valves-potential-risk-early-structural-valve-deterioration-letter-health-care (16 July 2024, date last accessed).

[2] David TE, Feindel CM, Bos J, Ivanov J, Armstrong S. Aortic valve replacement with Toronto SPV bioprosthesis: optimal patient survival but suboptimal valve durability. J Thorac Cardiovasc Surg 2008;135:19–24.

[3] Klautz RJM, Kappetein AP, Lange R, Dagenais F, Labrousse L, Bapat V et al; PERIGON Investigators. Safety, effectiveness and haemodynamic performance of a new stented aortic valve bioprosthesis. Eur J Cardiothorac Surg 2017;52:425–31.

[4] Sabik JF, Rao V, Lange R, Kappetein AP, Dagenais F, Labrousse L et al; PERIGON Investigators. One-year outcomes associated with a novel stented bovine pericardial aortic bioprosthesis. J Thorac Cardiovasc Surg 2018;156:1368–77.e5.

[5] Klautz RJM, Dagenais F, Reardon MJ, Lange R, Moront MG, Labrousse L et al Surgical aortic valve replacement with a stented pericardial bioprosthesis: 5-year outcomes. Eur J Cardiothorac Surg 2022;62:ezac374.

[6] Akins CW, Miller DC, Turina MI, Kouchoukos NT, Blackstone EH, Grunkemeier GL et al Guidelines for reporting mortality and morbidity after cardiac valve interventions. Eur J Cardiothorac Surg 2008;33:523–8.

[7] International Standards Organization. Cardiovascular implants; cardiac valve prostheses. Part 2: surgically implanted heart valve substitutes (EN ISO 5840-2:2009). Geneva, Switzerland, 2009.

[8] Généreux P, Piazza N, Alu MC, Nazif T, Hahn RT, Pibarot P et al; VARC-3 Writing Committee. Valve Academic Research Consortium 3: updated endpoint definitions for aortic valve clinical research. J Am Coll Cardiol 2021;77:2717–46.

[9] Velders BJJ, Vriesendorp MD, Asch FM, Reardon MJ, Dagenais F, Moront MG et al Current definitions of hemodynamic structural valve deterioration after bioprosthetic aortic valve replacement lack consistency. JTCVS Open 2024;19:68–90.

[10] Beaver T, Bavaria JE, Griffith B, Svensson LG, Pibarot P, Borger MA et al; COMMENCE Trial Investigators. Seven-year outcomes following aortic valve replacement with a novel tissue bioprosthesis. J Thorac Cardiovasc Surg 2024;168:781–91.

[11] Tsui S, Rosenbloom M, Abel J *et al.* Eight-year outcomes of aortic valve replacement with the Carpentier-Edwards PERIMOUNT Magna Ease valve. J Card Surg 2022;37:4999–5010.

[12] Werner P, Coti I, Kaider A, Gritsch J, Mach M, Kocher A et al Long-term durability after surgical aortic valve replacement with the Trifecta and the Intuity valve: a comparative analysis. Eur J Cardiothorac Surg 2022; 61:416–24.

[13] Ius F, Schulz J, Roumieh M, Fleissner F, Ismail I, Tudorache I et al Long-term results of the Mitroflow aortic pericardial bioprosthesis in over 800 patients: limited durability and mechanisms of dysfunction. Eur J Cardiothorac Surg 2017;52:264–71.

[14] Zoghbi WA, Chambers JB, Dumesnil JG, Foster E, Gottdiener JS, Grayburn PA et al; Canadian Society of Echocardiography. Recommendations for evaluation of prosthetic valves with echocardiography and Doppler ultrasound: a report from the American Society of Echocardiography's Guidelines and Standards Committee and the Task Force on Prosthetic Valves, developed in conjunction with the

American College of Cardiology Cardiovascular Imaging Committee, Cardiac Imaging Committee of the American Heart Association, the European Association of Echocardiography, a registered branch of the European Society of Cardiology, the Japanese Society of Echocardiography and the Canadian Society of Echocardiography, endorsed by the American College of Cardiology Foundation, American Heart Association, European Association of Echocardiography, a registered branch of the European Society of Cardiology, the Japanese Society of Echocardiography, and Canadian Society of Echocardiography. J Am Soc Echocardiogr 2009;22:975–1014.

[15] Lancellotti P, Pibarot P, Chambers J, Edvardsen T, Delgado V, Dulgheru R et al Recommendations for the imaging assessment of prosthetic heart valves: a report from the European Association of Cardiovascular Imaging endorsed by the Chinese Society of Echocardiography, the Inter-American Society of Echocardiography, and the Brazilian Department of Cardiovascular Imaging. Eur Heart J Cardiovasc Imaging 2016;17:589–90.

[16] Van Mieghem NM, Popma J, Sondergaard L, Oh J, Chang Y, Reardon M. Clinical outcomes and valve hemodynamics following transcatheter and surgical aortic valve replacement. In: Cardiovascular Research Technologies Conference, National Harbor, MD, USA, 22–25 February 2020.

[17] Hahn RT, Douglas PS, Jaber WA, Leipsic J, Kapadia S, Thourani VH et al Doppler velocity index outcomes following surgical or transcatheter aortic valve replacement in the PARTNER trials. JACC Cardiovasc Interv 2021;14:1594–606.

© The Author(s) 2024. Published by Oxford University Press on behalf of the European Association for Cardio-Thoracic Surgery.
This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial License (https://creativecommons.org/licenses/by-nc/4.0/), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited. For commercial re-use, please contact journals.permissions@oup.com
European Journal of Cardio-Thoracic Surgery, 2024, 67, 1–8
https://doi.org/10.1093/ejcts/ezae414
Original article